

# MANAN RASTOGI

AI Product Engineer | Applied CV, LLM Evaluation, Full-Stack Systems

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## EDUCATION

**Vellore Institute of Technology (VIT)**

*B.Tech in Electronics & Communication Engineering (Biomedical Specialization)*

Vellore, India

**2024 – PRESENT**

- **Core:** Digital Signal Processing, Biomedical Imaging, Neural Networks, Embedded Systems, Microprocessors & Micro-controllers.

**IIT Madras**

*BS in Data Science & Applications*

Online, India

**2025 – PRESENT**

- **Core:** Machine Learning Foundations, Statistical Modeling, Algorithmic Design, Business Data Analytics.

## TECHNICAL EXPERTISE

**Applied AI & Evaluation** Python 3.12, LLM APIs (NVIDIA LLaMA 3.1), prompt engineering, RAG pipelines, rule-based validation, OpenCV, scikit-image, TrackPy, SciPy, NumPy.

**Web & Full-Stack** TypeScript, React.js, Next.js, Node.js, Three.js (R3F), Framer Motion, Flask, RESTful APIs, Tailwind CSS, MongoDB, Vite.

**Systems & Tooling** Git, Docker, Vercel, MongoDB Atlas, SQLite, Tesseract OCR, CI/CD, Linux Bash, performance profiling.

**ECE & Embedded** Verilog, C++, digital logic, DSP fundamentals, embedded systems, microcontrollers.

## TECHNICAL PROJECT DEPLOYMENTS

**WoundTrack AI: Applied Computer Vision Pipeline**

**2024 – PRESENT**

*Python, OpenCV, Flask, SciPy, SQLite*

[\[Live Demo\]](#) [\[Source\]](#)

- Engineered a research-grade CV pipeline tracking **441 cells** across **800+ frames** in **<10 min** on consumer hardware, achieving **28% higher precision** than manual ImageJ workflows.
- Implemented LoG blob detection + LAP gap-closing tracker with Kalman prediction for explainable cell localization and trajectory continuity through missed frames.
- Designed custom evaluation metrics (MSD exponents, wavefront roughness, fractal-dimension analysis) to validate output against biological movement patterns.
- Built Flask + SQLite dashboard for batch processing, kinetic visualization, and automated report generation.

**VITalize AI: Full-Stack Academic Platform**

**2025 – PRESENT**

*Next.js 15, TypeScript, MongoDB, NVIDIA LLaMA 3.1, Tesseract OCR*

[\[Live Demo\]](#) [\[Source\]](#)

- Built a unified academic platform indexing **3,000+** past exam papers with full-text search, subject-level filters, and bulk download — replacing scattered Telegram groups and Google Drive folders for VIT's student body.
- Integrated **NVIDIA LLaMA 3.1** API with Tesseract OCR fallback pipeline to extract text from scanned PDFs and generate unit-wise, mark-weighted AI solutions with KaTeX rendering.
- Engineered constraint-based **FFCS timetable solver** with clash elimination, faculty prioritization, gap minimization, and ICS calendar export.
- Deployed on **Vercel** with **MongoDB Atlas**; built a Chrome extension for VTOP data sync; open-source with community funding.

**Interactive 3D Engineering Portfolio**

**2024 – PRESENT**

*TypeScript, React, Three.js (R3F), Framer Motion, Tailwind CSS*

[\[Live Demo\]](#) [\[Source\]](#)

- Built a portfolio with **React Three Fiber** 3D background scene (floating wireframe geometries, WebGL star fields) and **Framer Motion** scroll-triggered section reveals.
- Achieved **WCAG accessibility** with skip navigation, keyboard-dismissible intro, ARIA labels, live regions, and reduced-motion support.
- Responsive design with lazy-loaded 3D scene, chunk-split bundling, and scroll progress indicator.

**Editron — Open Source Contributor (GSSoC '26)**

**2026**

*JavaScript, Git, Open Source Workflow*

- Contributing code-quality improvements to Editron, an open-source code editor, through GirlScript Summer of Code 2026.
- Following standard open-source workflow: issue triage, fork, feature branch, and pull request.